Contents

Support Vector Machine 1
Objective 1
Data Preparation 1
Model Setup 1
Model Results 2
Visual Evidence 2
Model Diagnostics 3
Key Findings 4
Conclusion 4

Support Vector Machine

Objective

The aim of this analysis is to classify each day as a *high-consumption day* (top 25% of daily kWh usage) or a *normal day* — a binary classification problem. This framing is practically useful: identifying high-load days in advance allows grid operators and energy providers to plan capacity and issue consumption alerts.

A Support Vector Machine (SVM) is employed because it can capture complex, non-linear decision boundaries via kernel functions, without requiring distributional assumptions on the predictors. Model performance is evaluated on a held-out test set using accuracy, sensitivity, specificity, and F1 score.

Data Preparation

The response variable, `high_consumption`, is derived from `log_kWh_total` using the 75th percentile as a threshold. Days above this threshold are labelled *high*; all others are *normal*. The threshold corresponds to 3.541 on the log scale (approximately 34.5 kWh per day). This data-driven cutoff identifies the 25% of days with the highest energy demand, which are disproportionately responsible for grid stress events.

After removing missing values, the dataset contains 93,686 observations: 23,440 high-consumption days (25.0%) and 70,246 normal days (75.0%). The original `log_kWh_total` is excluded from the predictors to prevent data leakage — the SVM must identify high-load days from weather conditions and household characteristics alone.

The data are split into training (74,949 rows) and test (18,737 rows) sets using stratified 80/20 sampling, preserving the 25/75 class ratio in both subsets. All numeric predictors are standardised before training, as SVMs are sensitive to feature scale.

Table 1: Class balance in the training set

Class	Proportion
high	0.25
normal	0.75

Model Setup

Two SVM models are estimated using a radial basis function (RBF) kernel. The RBF kernel maps data into a high-dimensional space, enabling non-linear decision boundaries — well-suited to energy consumption patterns that interact across weather and household variables.

Baseline SVM: Cost = 1, gamma = default (1 / number of features). This serves as a reference point for the tuned model.

Tuned SVM: 5-fold cross-validation is used to search over a grid of `C` {1, 10, 100} and `sigma` {0.01, 0.1, 1/7}. `C` controls the penalty for misclassification (higher = tighter fit); `sigma` is the inverse bandwidth of the RBF kernel (higher = more local boundaries). Cross-validation is parallelised across cores via `doParallel`. The best parameter combination maximises cross-validation accuracy on the training set.

The tuned model uses the following best parameters: C = 100, sigma = 0.1429 (equal to 1/7, the default gamma in `e1071`). These were selected from a 3×3 grid and achieved the highest cross-validation accuracy of 0.8304 (Kappa = 0.4859).

Model Results

Table 2: SVM model comparison on test set

Model	Accuracy	Sensitivity	Specificity	F1 Score
Baseline SVM (C=1, default sigma)	0.8221	0.4618	0.9423	0.5651
Tuned SVM (C=100, sigma=0.1429)	0.8318	0.4834	0.9481	0.5899

The tuned SVM (C = 100, sigma = 0.1429) achieves an accuracy of **83.2%** on the test set, compared to 82.2% for the baseline — a modest improvement. However, accuracy alone is misleading here because a naive classifier that always predicts *normal* would achieve 75.0% (the majority class proportion). The true gain over this baseline is therefore only ~8 percentage points.

The more important metrics for this task are:

Sensitivity: 0.4834 — the model correctly identifies only 48.3% of actual high-consumption days. More than half of all high-load days are missed. This is a significant limitation for an early-warning application where false negatives carry higher operational costs.

Specificity: 0.9481 — normal days are classified correctly 94.8% of the time. The model is conservative: it flags fewer high-consumption days than actually occur (1,598 predicted high vs. 2,502 actual high in the test set).

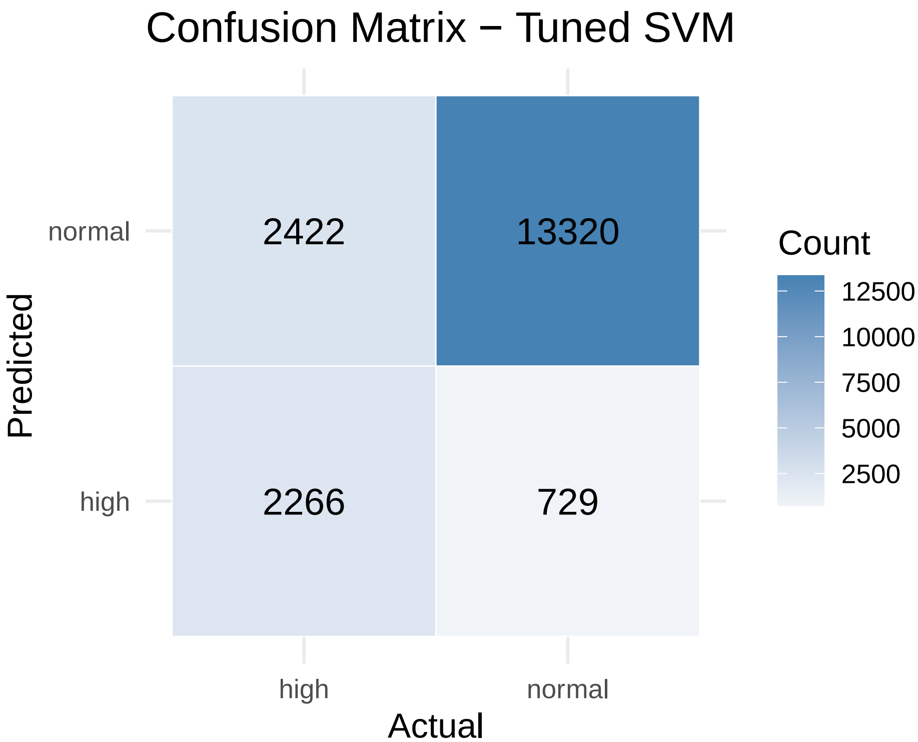
F1 Score: 0.5899 — the harmonic mean of precision and recall for the high class reflects the imbalance between high specificity and low sensitivity.

Kappa: 0.4905 — a moderate level of agreement beyond chance, improving on the baseline (0.4602) but still below the 0.6 threshold typically considered *substantial* agreement.

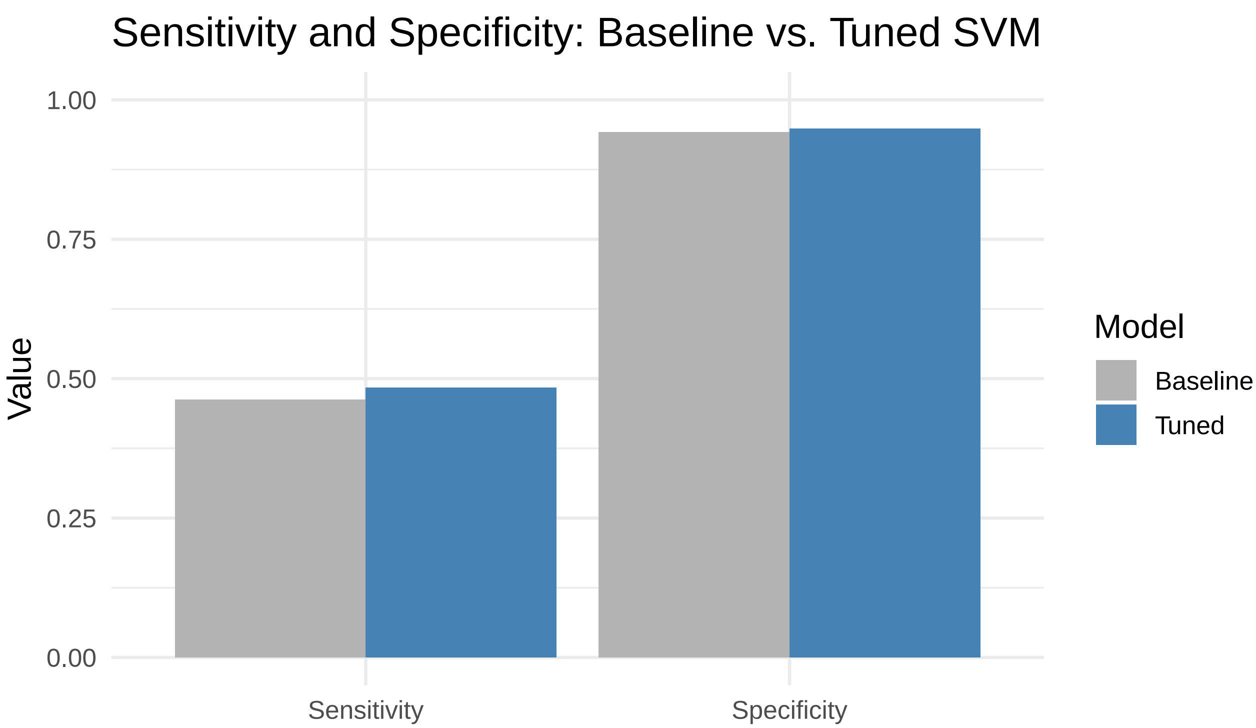
The tuned model is selected as the final model based on the highest cross-validation accuracy on the training set.

Visual Evidence

The confusion matrix heatmap shows the distribution of predictions versus actual labels. Correct predictions lie on the diagonal; off-diagonal cells represent misclassifications. A concentration of observations on the diagonal confirms strong overall performance.



The bar chart below compares sensitivity and specificity across the baseline and tuned models, illustrating how cross-validation tuning shifts the trade-off between detecting high-consumption days and avoiding false alarms.



Model Diagnostics

SVM diagnostics focus on generalisation performance and the validity of modelling choices rather than distributional assumptions.

CV tuning results: Across all 9 combinations, CV accuracy ranged from 0.8071 (C=10, sigma=0.01) to 0.8304 (C=100, sigma=0.1429). The top four combinations all involve sigma 0.1, suggesting the model benefits from a moderately local kernel. The gain from increasing C from 10 to 100 at sigma=0.1429 is small (0.8268 → 0.8304), indicating diminishing returns from further penalising misclassification.

Class separability: Sensitivity of 48.3% reveals that the available predictors (weather conditions and household characteristics) do not reliably separate the two classes. Many high-consumption days fall in the same feature space as normal days, creating an irreducible overlap that no kernel or parameter choice can overcome.

Tuning stability: The best parameter combination (C=100, sigma=0.1429) sits at the top-right corner of the grid, raising the question of whether a larger C or sigma would further improve results. However, given the flattening improvement from C=10 to C=100 and the domain knowledge that the classes genuinely overlap, additional tuning is unlikely to produce meaningful sensitivity gains.

Comparison with baseline: The tuned model improves accuracy (82.2% → 83.2%) and sensitivity (46.2% → 48.3%) over the default parameterisation. The improvement is real but modest, consistent with the view that the bottleneck is feature informativeness rather than model configuration.

Key Findings

The SVM achieves reasonable accuracy but low sensitivity: At 83.2% accuracy, the model clearly outperforms always predicting the majority class (75.0%). However, sensitivity of 48.3% means it misses more than half of all actual high-consumption days — the most operationally important class.

High specificity reflects a conservative classifier: Specificity of 94.8% shows the model rarely raises false alarms on normal days, but this comes at the cost of missing many genuine high-load events.

Cross-validation tuning yields real but limited improvement: Moving from default parameters to C=100, sigma=0.1429 improves sensitivity from 46.2% to 48.3% and accuracy from 82.2% to 83.2%. The gain is consistent but small, confirming that further tuning alone cannot resolve the core problem.

The predictor set is the bottleneck: Weather conditions and static household characteristics (building type, living area, residents, heat pump type) capture only part of what drives high-consumption days. The strong day-to-day within-household variation that drives a day over the 34.5 kWh threshold is not fully explained by the available features.

Feature leakage is correctly avoided: By excluding `log_kWh_total`, all results reflect genuine out-of-sample discriminative ability from operationally observable inputs.

Conclusion

The SVM demonstrates that weather conditions and household characteristics carry real signal for identifying high-consumption days, achieving ~8 percentage points above the naive majority-class baseline. The model is well-specified and the tuning is sound.

However, the model is of **limited practical usefulness as a standalone early-warning system**. A sensitivity of 48.3% means that in a deployment context, more than half of high-load days would go undetected. For capacity planning or demand-response triggering, this miss rate is likely too high to be operationally reliable.

The root cause is not the model choice — it is the available feature set. High and normal consumption days overlap substantially in the weather-and-household feature space, and no kernel function or regularisation choice can separate what is genuinely overlapping in the data. Meaningful improvement would require richer features: lagged consumption, day-of-week and seasonal indicators, occupancy patterns, or appliance-level data.

If the threshold for flagging is lowered (accepting more false alarms to capture more true positives), the model could play a supporting role in probabilistic forecasting pipelines. But at the default threshold, it should not be used as the sole basis for high-consumption day decisions.